

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Linear Algebra	Course Code:	MT1004
	Program:	BDS	Semester:	Fall 2021
	Duration:	15 Minutes	Total Marks:	10
	Paper Date:	November 25, 2021.	Weight	
	Section:	1B	Page(s):	
	Exam Type:	Quiz-II		

Student : Name: _____ Roll No. _____ Section: _____

Instruction/Notes: Programmable calculators are not allowed.

Question: For what value of λ do the following vectors form a linearly dependent set in $\mathbb{R}^3$?

$$v_1 = \left(\lambda, -\frac{1}{2}, -\frac{1}{2} \right), \quad v_2 = \left(-\frac{1}{2}, \lambda, -\frac{1}{2} \right), \quad v_3 = \left(-\frac{1}{2}, -\frac{1}{2}, \lambda \right)$$

Solution: Suppose that the given vectors are linearly independent, then they should satisfy the following condition

$$0 = k_1 v_1 + k_2 v_2 + k_3 v_3 \text{ ----- (1)}$$

And has at least one solution, namely $k_1 = 0, k_2 = 0, k_3 = 0$ otherwise dependent. _____ 3

In other words, if the corresponding homogeneous system has unique solution, then it is independent otherwise dependent.

The coefficient matrix of the corresponding system is $A = \begin{bmatrix} \lambda & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \lambda & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \lambda \end{bmatrix}$ _____ 3

For infinite solution, the determinant of the A should be zero. i.e. $|A| = 0$

$$\begin{vmatrix} \lambda & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \lambda & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \lambda \end{vmatrix} = 0 \quad \Rightarrow \quad 8\lambda^3 - 6\lambda - 2 = 0$$

$$\Rightarrow \quad \boxed{\lambda = 1, -\frac{1}{2}}$$

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Question: Show that the vectors $v_1 = (1, 2, 3, 4)$, $v_2 = (0, 1, 0, -1)$, and $v_3 = (1, 3, 3, 3)$ form a linearly dependent set in $\mathbb{R}^4$. Also express each vector as a linear combination of the other two.

Solution: Suppose that the given vectors are linearly independent, then they should satisfy the following condition

$$0 = k_1 v_1 + k_2 v_2 + k_3 v_3 \text{ ----- (1)}$$

$$(0, 0, 0, 0) = k_1 (1, 2, 3, 4) + k_2 (0, 1, 0, -1) + k_3 (1, 3, 3, 3)$$

Form the linear combination, we get the following system

$$k_1 + 0k_2 + k_3 = 0$$

$$2k_1 + 1k_2 + 3k_3 = 0$$

$$3k_1 + 0k_2 + 3k_3 = 0$$

$$4k_1 - k_2 + 3k_3 = 0$$

The reduced row echelon form of the above system is

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now solution is $k_1 = -s, \quad k_2 = -s, \quad k_3 = s$

The solution of the above system of equation is non-trivial, i.e. the system possesses infinite many solutions.

Therefore, the given set of vectors are linearly dependent.

Now to express the linear combination, we assume that the value of parameter $s = 1$,

Equation 1 reduces to $0 = -v_1 - v_2 + v_3$

$$v_3 = v_1 + v_2$$